

Hands-On Activity 6

Assume that we have the following API for the Binary Tree, for the questions below.

```
public class BinaryTree {

    private Node root;

    private class Node {
        private int key;           % key
        private Node left;        % left child
        private Node right;       % right child
    }

}
```

1. Write a **recursive** Java solution to find the minimum key in the Binary Tree. The method should be implemented as a method of the `BinaryTree` class.
2. Write a **recursive** Java solution to find the total number of the keys in the Binary Tree. The method should be implemented as a method of the `BinaryTree` class.

3. Write a **recursive** Java solution to find the height of the Binary Tree. The method should be implemented as a method of the `BinaryTree` class.

4. Write a **recursive** Java solution to find the sum of all the keys in the Binary Tree. The method should be implemented as a method of the `BinaryTree` class.